

AES Project Tracker & Management System

1. Introduction

AES Project Tracker & Management System, is a centralized web-based platform designed to manage projects, team members, clients, tickets, communications, approvals, and project progress from initiation to completion.

The system will function as a project management and client collaboration platform similar to Jira, customized specifically for AES workflows and organizational requirements.

2. Objectives

The primary objectives of the system are:

- Centralize project management activities.
- Manage multiple projects simultaneously.
- Improve communication between AES and clients.
- Track project progress in real-time.
- Manage tickets, feature requests, and bug reports.
- Assign and monitor developer and intern activities.
- Maintain project documentation.
- Improve accountability through activity tracking.
- Generate reports and project analytics.

3. User Roles

The system shall support the following user roles:

3.1 Admin

The Admin is the highest authority in the system.

Responsibilities

- Create and manage users.
- Assign user roles.
- Create and manage projects.
- Assign team members to projects.
- Manage project status.
- Review and approve feature requests.

- Manage billing and commercial discussions.
- Generate reports.
- Monitor overall project activities.

3.2 Developers

Responsibilities

- View assigned projects.
- View assigned tickets and tasks.
- Update ticket status.
- Add comments and progress updates.
- Upload deliverables.
- Escalate issues when required.
- Request commercial review if a ticket is incorrectly categorized.

3.3 Interns

Responsibilities

- View assigned projects.
- Work on assigned tasks.
- Update progress.
- Upload completed work.
- Participate in project discussions.

Intern permissions may be limited compared to developers.

3.4 Clients

Responsibilities

- View project status.
- Raise support tickets.
- Submit feature requests.
- Report bugs.
- Participate in project discussions.
- Upload supporting documents.
- Review completed work.
- Join project meetings.

4. Authentication and User Management

User Authentication

The system shall support:

Phase 1

- Google Authentication (Google Sign-In)

Future Enhancements

- Email and Password Login
- Google Authentication
- Microsoft Authentication

User Creation

Admins shall be able to:

- Create users.
- Edit users.
- Deactivate users.
- Reset passwords.
- Assign roles.

User Information

- Name
- Email Address
- Phone Number
- Role
- Designation
- Organization
- Status (Active/Inactive)

5. Project Management Module

Project Creation

The Admin shall be able to create new projects.

Project Information

- Project Name
- Project Code
- Client Name
- Organization Name
- Project Description
- Project Type
- Technology Stack
- Start Date
- Expected End Date
- Priority
- Status

Project Statuses

The system shall support the following project statuses:

- Proposal Received
- In Progress
- Maintenance
- On Hold
- Cancelled
- Completed

6. Project Team Mapping

Each project may have multiple associated users.

Example

Project A

- 1 Admin
- Multiple Developers
- Multiple Interns
- Multiple Client Representatives

The Admin shall be able to add, remove, and modify project members at any time.

Users shall only be able to view projects to which they are assigned. Clients shall only have access to their own projects and related tickets.

7. Ticket Management Module

The Ticket Management Module is the core component of the system.

Clients shall be able to create tickets related to their projects.

Ticket Categories

The system shall support the following ticket types:

Bug Fix

Issues related to existing functionalities.

New Feature Request

Requests for additional functionality.

Enhancement Request

Improvements to existing features.

Technical Support

General support requests.

Ticket Information

Each ticket shall contain:

- Ticket ID
- Title
- Description
- Category
- Priority
- Associated Project
- Attachments
- Created By
- Assigned To
- Creation Date
- Due Date
- Current Status

8. Ticket Workflow

Bug Fix Workflow

Client

- Ticket Creation
- Developer Assignment
- Development
- Testing
- Resolution
- Client Notification

New Feature Workflow

Client

- Feature Request
- Admin Review
- Cost Estimation
- Client Approval
- Payment Approval
- Development Team Assignment
- Development
- Testing
- Deployment
- Completion

Commercial Review Workflow

Sometimes clients may classify a new feature as a bug fix.

In such cases:

Developer

- Identifies Incorrect Category
- Sends Ticket for Commercial Review
- Admin Reviews Requirement
- Cost Estimation
- Client Approval
- Development Process

9. Ticket Statuses

The system shall support the following ticket statuses:

- Open
- Awaiting Admin Approval
- Awaiting Payment
- Approved
- In Development
- Resolved
- Reopened
- Closed
- Rejected
- On Hold

The system shall allow users to upload supporting attachments while creating or updating tickets.

10. Task Management

A ticket may be divided into multiple smaller tasks.

- Task Name
- Assigned Member
- Due Date
- Status

11. Email Notification System

The system shall automatically send email notifications for:

Project Events

- Project Creation
- Team Member Assignment
- Status Updates

Ticket Events

- Ticket Creation
- Ticket Assignment
- Ticket Status Changes
- Ticket Completion

Commercial Events

- Cost Approval Required
- Payment Pending
- Payment Approved

Meeting Events

- Meeting Scheduled
- Meeting Reminder

12. Meeting Management Module

The system shall support scheduling project-related meetings.

Meeting Details

- Meeting Title
- Agenda
- Date
- Time
- Participants
- Meeting Link

Meeting invitations shall be automatically emailed to all project participants.

13. Document Management Module

Each project shall maintain a centralized document repository for storing project-related files and documentation managed by the administrator.

14. Dashboard Module

Admin Dashboard

The dashboard shall display:

- Total Projects
- Active Projects
- Completed Projects
- Open Tickets
- Closed Tickets
- Revenue Generated

Developer Dashboard

The dashboard shall display:

- Assigned Projects
- Assigned Tickets

Client Dashboard

The dashboard shall display:

- Active Projects
- Open Tickets

15. Reports and Analytics

The system shall generate reports for:

Project Reports

- Project Progress
- Project Completion Status
- Delayed Projects

Ticket Reports

- Open Tickets
- Closed Tickets

Reports shall be exportable to PDF and Excel formats.

16. Calendar Module

The system shall provide a calendar view displaying:

- Project Deadlines
- Task Deadlines
- Meetings
- Deliverables
- Milestones

Calendar Integration

The system shall support synchronization with external calendar platforms, including like Google Calendar or similar.

Users shall be able to:

- Add project events directly to their personal calendar
- Automatically sync project deadlines and meetings
- Receive calendar reminders